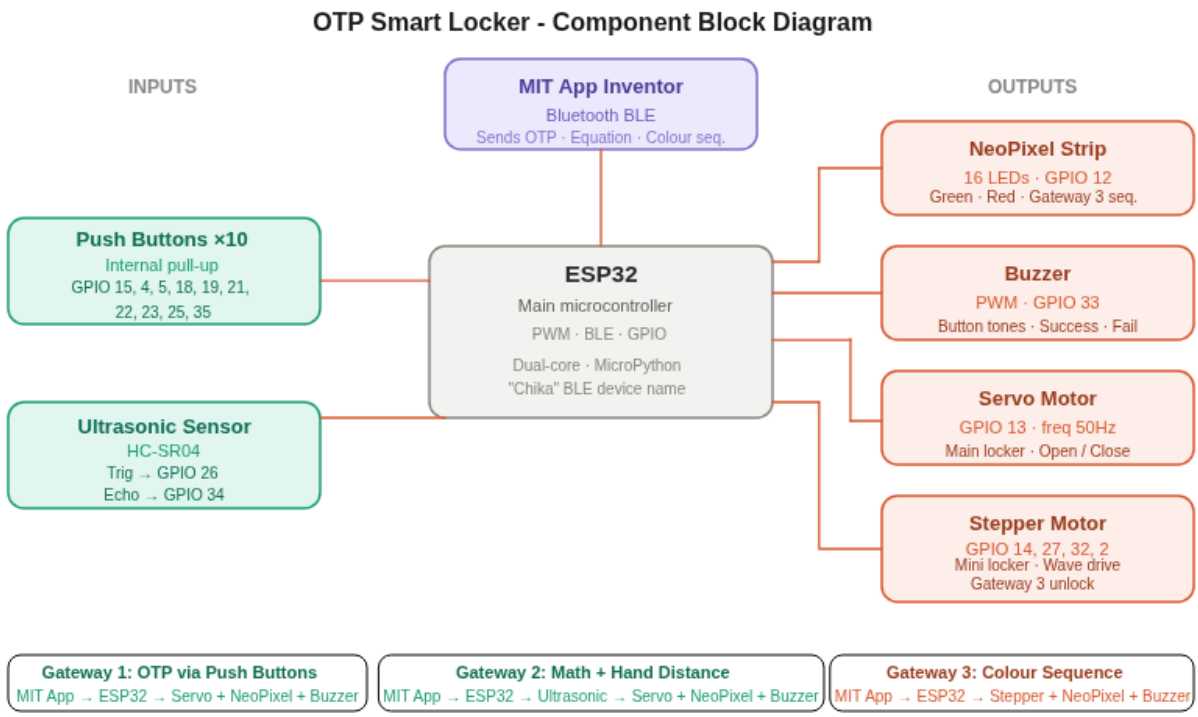
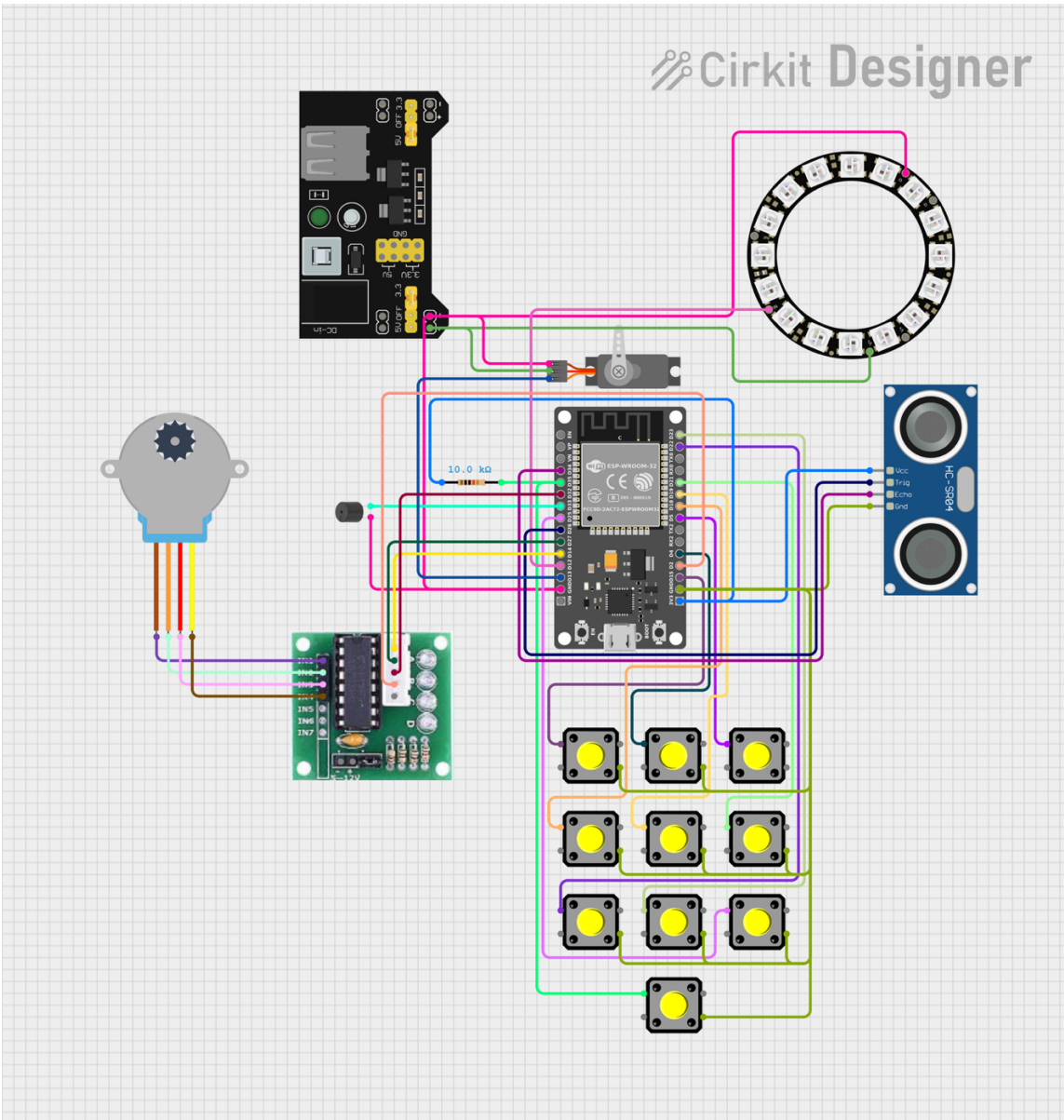
Electronics Documentation

1. Electronics Block Diagram



2. Components Table

Component	Quantity	Purpose	Justification
INPUT			
Pushbuttons	10	Physical keypad for OTP entry.	Provides tactile, physical interaction for the user to input the generated code.
Ultrasound Sensor [HC-SR04]	1	Measures hand distance between the user and the safe [math equation]	Allows for touchless interaction and precise distance measurements (evaluated with a +/- 2.0cm tolerance).
PROCESSING			
ESP32	1	• Receives input (from BLE, pushbuttons and HC-SR04) • Receives and executes system logic	Required for built-in BLE capabilities to communicate with the app, plus the GPIO pins

		• Controls Hardware (servo, stepper, NeoPixel, buzzer)	
OUTPUT			
Servo motor	1	Operates the main sliding door mechanism	• Provides exact positional control (110 to 70 degrees) to reliably open and close the main door. • Most preferable as our main door had a linear sliding mechanism (using a scissor structure). • A quick momentum was needed and its small size made it easier to position it below the door.
Stepper motor	1	Operates the mini locker mechanism	• Wave-drive stepping allows for accurate rotational control. • Preferable as its light (was required to be stuck on the wall). • Anti-clockwise, clockwise rotation.
NeoPixel	1	Visual feedback and custom color sequences • Correct sequence – green • Wrong sequence – red • Mini locker – multicolour	Requires only one data pin to control multiple LEDs, saving crucial GPIO pins on the ESP32.
Buzzer	1	Auditory feedback • Different frequency for each pushbutton when OTP is entered. • Correct sequence • Wrong sequence • Opening of main locker and mini locker	Crucial for accessibility and user feedback
POWER			
SMPS Adaptor	1	Provides voltage to power module	To convert wall AC power into a stable, high-current 5V DC power source for the entire circuit.
Power Module	1	• Distributes power to high-draw components • Provides voltage to NeoPixel, servo and stepper motor	The ESP32 cannot safely output enough current to drive the motors and NeoPixel simultaneously; external power prevents brownouts.

Motor Driver	1	Controls the stepper motor	To adjust the voltage of the coils within the stepper motor.
EXTRA			
Jumper wires (Male and Female)	Many.	Connection of all the components	Makes tighter connections, easier to tape up as compared to normal wire.
10k ohm resistor	1	External pull-up resistor for Pin 35 (pushbutton)	Pin 35 is input-only and lacks an internal pull-up resistor; required to prevent floating signals and "ghost typing."